

Expected Returns

MSCF Investments

Fall 2021 Mini 1

- ❑ Risk/Reward trade-off
 - CAPM
 - Arbitrage Pricing Theory (multi-factor risk and reward)
- ❑ Financial Economics
 - What makes a random variable priced?
- ❑ Cross-section of expected returns
- ❑ Fama-MacBeth estimation
- ❑ Long-short portfolios
- ❑ Strategy back-testing
- ❑ Examples
 - Is liquidity priced?
 - Haugen and Baker “kitchen sink” predictive model

Introduction

□ **Expected return:** $E_{t-1}[r_{j,t}] = \frac{E_{t-1}[c_{j,t} + P_{j,t} - P_{j,t-1}]}{P_{j,t-1}}$

□ **Problem:**

Market determines prices P that induce investor beliefs about $E[\text{ret}]$ sufficient that markets clear given associated induced aggregate investor demand $D(P)$.

Investors see market price P , but expected returns are **not directly observable**. Rather, investors have **views about $E[\text{ret}]$** given market prices and their views about future cash flows and prices.

Thus, investors **model/estimate** expected returns as inputs in their investment problem.

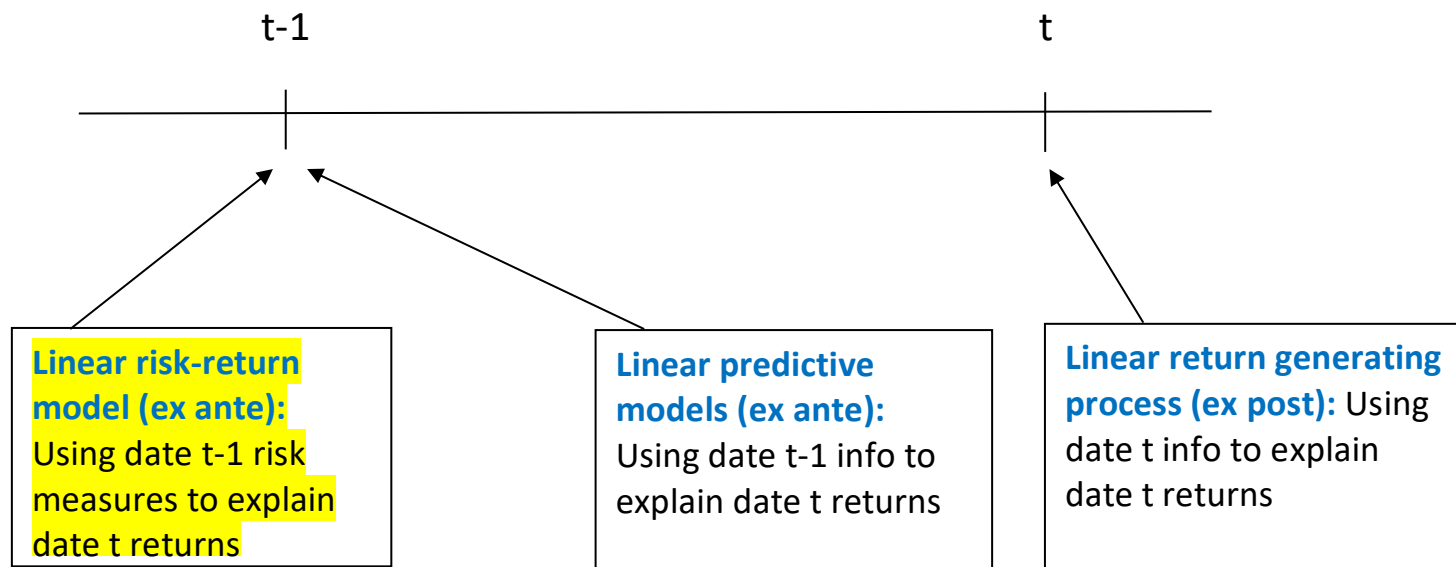
□ **Whose expectation?**

- Market expectation/market discount rate $E_{t-1}^m[r_{j,t}]$
- Investor expectation $E_{t-1}^{Inv}[r_{j,t}]$
- Statistical expectation $E_{t-1}[r_{j,t}]$.
- Potentially **different information**
- Potentially **different understanding** of **probabilistic structure** relating information to returns

□ **Question:** Conceptually, what drives an investor's views about expected returns?

Linear models again

Timing:



Linear models of risk/reward

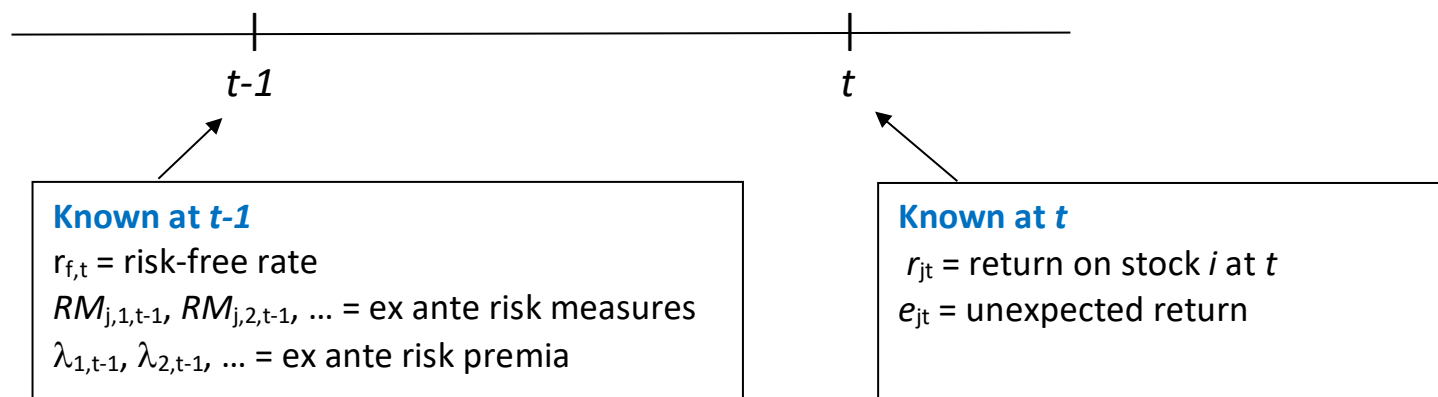
- Relate expected returns $E_{t-1}[r_{jt}]$ given date $t-1$ prices to risk measures known at $t-1$.

Linear predictive models

- Predicts returns r_{jt} at t using prior explanatory variables known at $t-1$.

Both models can be used for security selection.

Expected returns and linear risk/reward relation



□ Returns and expected returns

$r_{jt} = [\text{expected return given everything known at } t-1] + [\text{surprise at } t]$

$$= E_{t-1}[r_{jt}] + e_{jt}$$

$$= E[r_{jt} \mid \text{risk measures \& risk premia known at } t-1] + e_{jt}$$

$$= r_{f,t} + \lambda_{1,t-1} RM_{j,1,t-1} + \lambda_{2,t-1} RM_{j,2,t-1} + \dots + e_{jt}$$

Hypothesis: Expected return is just **compensation** for risk

Hypothesis: Expected return given risk-free rate $r_{f,t}$ is **linear** in **constant** risk measures $RM_{j,1,t-1}, RM_{j,2,t-1}, \dots$ and **constant** risk premia $\lambda_{1,t-1}, \lambda_{2,t-1}, \dots$

Example: Sharpe-Lintner Capital Asset Pricing Model (CAPM)

Securities Market Line: Cross-sectional asset pricing and risk-reward trade-off

$$E_{t-1}[r_{j,t}] = r_{f,t} + \beta_{j,m,t-1} (E_{t-1}[r_{m,t}] - r_{f,t})$$

- **Market beta** $\beta_{j,m,t-1} = \frac{cov_{t-1}(r_{j,t}, r_{m,t})}{var_{t-1}(r_{m,t})}$ measures asset j's **Market risk**.
- **Market risk premium** $\lambda_{m,t-1} = E_{t-1}[r_{m,t}] - r_{f,t}$. If non-zero, then market prices securities to reward bearing market risk – i.e., market risk is **priced**!

$$P_{j,t-1} = \frac{E_{t-1}[c_{j,t} + P_{j,t}]}{1 + E_{t-1}[r_{j,t} | \text{all characteristics of } j \text{ at time } t]} = \frac{E_{t-1}[c_{j,t} + P_{j,t}]}{1 + \boxed{r_{f,t} + \lambda_{m,t-1} \beta_{j,m,t-1}}}$$

- Strong prediction of CAPM: Market risk $\beta_{j,m,t-1}$ is **only** individual stock characteristic that is **priced**!
- Unpriced characteristics in CAPM? Everything else about stock j:
 - Idiosyncratic risk, other risk metrics, other Fama-French factor sensitivities, liquidity, size...
- **Question:** How do investor increase their expected returns in a CAPM world?

CAPM assumptions

- All investors have **mean-variance preferences**
- All investors **agree** on return means, SDs and correlations
- All investors can **borrow & lend** at risk-free rate $r_{f,t}$.
 - **Two-fund Separation**: All investors **hold same risky tangency portfolio** + borrowing/lending
 - **CAPM equilibrium**: **Tangency** portfolio = **Market** portfolio of risky assets

Comments

□ CAPM vs. Linear Factor Models of (ex post) returns

- Possible firms may have **multi-factor return-generating model**

$$r_{j,t} = \beta_{j,0} + \beta_{j,1}F_{1,t} + \dots + \beta_{j,k}F_{k,t} + \dots + \beta_{j,K}F_{K,t} + e_{j,t}$$

but still **only market risk $\beta_{j,m}$ is priced** in CAPM

□ Time-varying expected returns with conditional CAPM:

$$E_{t-1}[r_{j,t}] = r_{f,t} + \beta_{j,m,t-1} (E_{t-1}[r_{m,t}] - r_{f,t})$$

- Dynamic market exposures: Time-varying $\beta_{j,m,t-1}$
- Time-varying market risk premium: $\lambda_{m,t-1} = E_{t-1}[r_{m,t}] - r_{f,t}$
- **Question:** What could cause such time-variation?

□ CAPM and MV analysis are **related but different**.

- CAPM is a **theory of equilibrium market asset pricing** (i.e., expected returns)
- MV analysis is a **tool for optimal portfolio choice** for individual investors
- CAPM could be false but an individual MV investor might still do MV analysis

Broader definition of risk

- Suppose return-generating process is a Linear Factor Model

$$r_{j,t} = \beta_{j,0} + \beta_{j,1}F_{1,t} + \dots + \beta_{j,k}F_{k,t} + \dots + \beta_{j,K}F_{K,t} + e_{j,t}$$

where, suppose, factors $F_{1,t}, \dots, F_{K,t}$ account for **all** common co-movement in returns

$$\text{cov}(r_{j,t}, r_{i,t}) = \text{cov}(\beta_{j,0} + \beta_{j,1}F_{1,t} + \dots + \beta_{j,K}F_{K,t} + e_{j,t}, \beta_{i,0} + \beta_{i,1}F_{1,t} + \dots + \beta_{i,K}F_{K,t} + e_{i,t})$$

$$= \sum_{k=1}^K \beta_{j,k} \beta_{i,k} \text{var}[F_{k,t}] + \sum_{k=1}^K \sum_{z \neq k} \beta_{j,k} \beta_{i,z} \text{cov}[F_{k,t}, F_{z,t}]$$

- In other words, suppose $\text{cov}(e_{j,t}, e_{i,t}) = 0$ for all stocks j and k (i.e., residuals are **idiosyncratic**)
- In practice, hope a small number of factors K is enough.

- **What is “risk”** of stock j viewed as of time $t-1$ looking forward to time t ?

- $\beta_{j,k}$ = sensitivity of return $r_{j,t}$ to randomness induced by exposure to factor $F_{k,t}$ for $k = 1, \dots, K$
- $\text{SD}(e_{j,t})$ = amount of idiosyncratic randomness in $r_{j,t}$ due to j 's residual $e_{j,t}$

Arbitrage Pricing Theory

- Given a Linear Factor Model of returns

$$r_{j,t} = \beta_{j,0} + \beta_{j,1}F_{1,t} + \dots + \beta_{j,k}F_{k,t} + \dots + \beta_{j,K}F_{K,t} + e_{j,t}$$

the **APT expected return** on each stock j is

$$E_{t-1}[r_{j,t}] = \lambda_{0,t-1} + \lambda_{1,t-1} \beta_{j,1} + \dots + \lambda_{K,t-1} \beta_{j,K}$$

where

$\lambda_{0,t-1} = r_{f,t} = E[\text{ret}]$ on a security or portfolio with **no systematic** risk.

- Doesn't matter whether stock is completely risk-free or random with just idiosyncratic risk.

$\lambda_{k,t-1} = \text{risk premium}$ at time $t-1$ for bearing factor $F_{k,t}$ risk at time t .

- Additional $E[\text{ret}]$ earned per additional unit of $\beta_{j,k}$ exposure to factor $F_{k,t}$
- Possible **time-variation**: Risk premium $\lambda_{k,t-1}$ may change with time $t-1$

Arbitrage Pricing Theory (2)

Interpretation

- Model of **cross-sectional differences** in expected returns across securities as of time $t-1$.
- **Price of risk**. If $\lambda_{k,t-1} \neq 0$, then factor k is “priced” in that $\beta_{j,1}$ affects market discount rate

$$P_{j,t-1} = \frac{E_{t-1}[c_{j,t} + P_{j,t}]}{1 + E_{t-1}[r_{j,t} | \text{all characteristics of } j \text{ at time } t]} = \frac{E_{t-1}[c_{j,t} + P_{j,t}]}{1 + [r_{f,t} + \lambda_{1,t-1}\beta_{j,1,t-1} + \dots + \lambda_{K,t-1}\beta_{j,K,t-1}]}$$

Two predictions

- Rewards for risk-bearing are **linear** in amount of risk as measured by loadings $\beta_{j,k,t-1}$
- **No reward** for bearing **idiosyncratic** risk
 - All investors should hold “**well-diversified**” portfolios.
 - Consistency in equilibrium: No idiosyncratic risk since diversified away, so idiosyncratic risk is uncompensated. So, no incentive to hold undiversified portfolios.

Relation of APT to CAPM

□ APT and linear factor model

- Return generation in linear factor model if market risk is included:

$$r_{j,t} - r_{f,t} = \beta_{j,0} + \beta_{j,1} [r_{m,t} - r_{f,t}] + \beta_{j,2} F_{2,t} + \dots + \beta_{j,K} F_{K,t} + e_{j,t}$$

where first factor is $F_{1,t} = r_{m,t} - r_{f,t}$

- Version of APT if only market risk is priced:

$$E_{t-1}[r_{j,t}] = r_{f,t} + \lambda_{1,t-1}^{APT} \beta_{j,1}$$

□ CAPM and linear market model

- Market model

$$r_{j,t} = \beta_{j,0} + \beta_{j,m} r_{m,t} + e_{j,t}$$

used to estimate CAPM beta $\beta_{j,m} = \frac{\text{cov}(r_{j,t}, r_{m,t})}{\text{var}(r_{m,t})}$

$$E_{t-1}(r_{j,t}) = r_{f,t} + \beta_{j,m} [E_{t-1}[r_{m,t}] - r_{f,t}]$$

Question: Is $\beta_{j,1} = \beta_{j,m}$? Probably not. If not, then no reason for $\lambda_{1,t-1}^{APT} = E_{t-1}[r_{m,t}] - r_{f,t}$.

Financial Economics

Risk premia $\lambda_{k,t-1}$ can be positive, negative or zero for a factor depending on:

- Nature of factor F_{kt}
- Investor preferences

Investor's problem

$$\text{Max}_{q_{i,1,t-1}, \dots, q_{i,N,t-1}} u_{i,t-1}(c_{i,t-1} \mid x_{i,t-1}) + E_{i,t-1}[V_{i,t}(W_{i,t} \mid x_{i,t})]$$

- **Utility** $u_{i,t-1}$ for investor i at t-1 from investor **consumption** $c_{i,t-1}$ and given **other things** $x_{i,t-1}$ investor i cares about at time t-1
 - $c_{i,t-1} = W_{i,t-1} + \sum_{j=1}^N q_{i,j,t-1} P_{j,t-1}$ = investor i consumption at t-1
 - $W_{i,t-1}$ = **Wealth** of investor i at time t-1 from all sources (i.e., financial portfolio, labor income)
- **Value function** $V_{i,t}$ of investor i at time t given investor total **wealth** $W_{i,t}$ and **other things** $x_{i,t}$ investor i cares about at time t
 - $W_{i,t} = \sum_{j=1}^N q_{i,j,t-1} * (P_{j,t} + c_{j,t}) + L_{i,t}$ = Wealth of investor i at time t from all sources
 - $\sum_{j=1}^N q_{i,j,t-1} * (P_{j,t} + c_{j,t})$ = Incoming portfolio value at time t given i's holdings from time t-1
 - $L_{i,t}$ = Labor and non-investment income of investor i at time t
- **Decision variable** at time t-1: $q_{i,j,t-1}$ = holdings (# of shares) of stock $j = 1, \dots, N$

Optimal portfolio

- Investing as intertemporal substitution
- Investors indifferent between using stock j to shift wealth at time $t-1$ into random wealth at time t
- **First-order condition (FOC)** for optimal portfolio

$$u'_{i,t-1} P_{j,t-1} = E_{t-1}[V'_{i,t} * (P_{j,t} + c_{j,t})] \quad \leftarrow \text{Optimal holdings } q_{i,j,t-1} \text{ chosen so that FOC holds}$$

$u'_{i,t-1}$ = investor i 's marginal utility of an additional dollar of consumption at time $t-1$

$V'_{i,t}$ = investor i 's marginal value of wealth at time t . Random at time $t-1$.

- FOC rewritten to solve for date $t-1$ prices:

$$\rightarrow P_{jt-1} = E_{t-1}[(V'_{i,t} / u'_{i,t-1}) * (P_{jt} + c_{jt})]$$

$$\rightarrow P_{jt-1} = E_{t-1}[\text{MRS}_{i,t-1,t} * (P_{jt} + c_{jt})]$$

$\text{MRS}_{i,t-1,t} = V'_{i,t} / u'_{i,t-1} = \text{marginal rate of substitution}$ between time $t-1$ and wealth at time t .

Strong implication: In equilibrium, pricing equation must hold for MRS's for each optimizing investor i

Why are some factors priced?

- Math reminder: $E(xy) = E(x)E(y) + \text{cov}(x, y)$

- Using this gives

$$P_{j,t-1} = E_{t-1}[MRS_{i,t-1,t} (P_{j,t} + c_{j,t})]$$

$$\rightarrow P_{j,t-1} = E_{t-1}[MRS_{i,t-1,t}] E_{t-1}[P_{j,t} + c_{j,t}] + \text{cov}_{t-1}(MRS_{i,t-1,t}, P_{j,t} + c_{j,t}).$$

- Note:** $E_{t-1}[MRS_{i,t-1,t}] = D_{t-1}$ = price of **1-period riskless bond**. Thus, from margin investor FOC

$$P_{j,t-1} = E_{t-1}[P_{j,t} + c_{j,t}] D_{t-1} + \text{cov}_{t-1}(MRS_{i,t-1,t}, P_{j,t} + c_{j,t}).$$

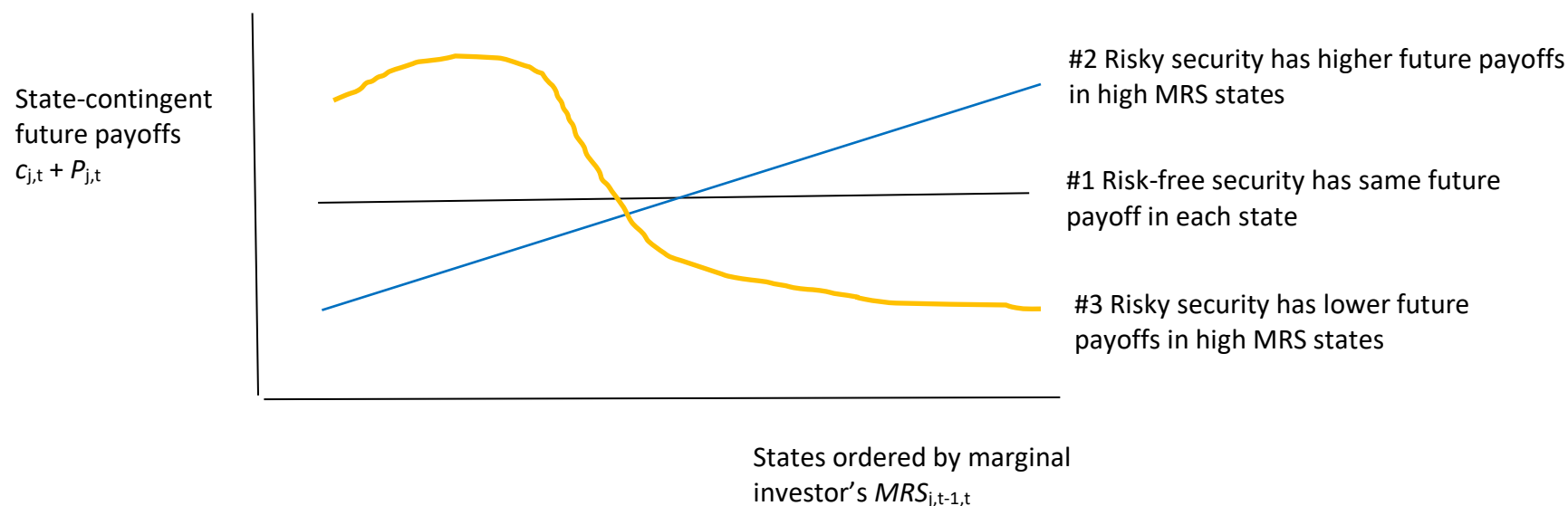
- Comparison with MDF formulation:** $P_{j,t-1} = E_{t-1}[c_{j,t} + P_{j,t}] MDF_{j,t-1}$

- Pricing:** If $\text{cov}_{t-1}(MRS_{i,t-1,t}, P_{j,t} + c_{j,t}) \neq 0$, then $MDF_{j,t-1} \neq D_{t-1} = \frac{1}{1 + r_{f,t-1}}$.

- Punch line:** For randomness in $P_{j,t} + c_{j,t}$ to be priced, it must covary with $MRS_{it} = V'_{i,t} / u'_{i,t-1}$ for marginal investor where $V'_{i,t} = \frac{dV_{i,t}(W_{i,t} | x_{i,t})}{dW_{i,t}}$ and $u'_{i,t-1} = \frac{du_{i,t-1}(c_{i,t-1} | x_{i,t-1})}{dc_{i,t-1}}$

Another covariance intuition

- In Asset Pricing slides, considered $\text{cov}_{t-1}(MDF_{j,t}, E_t[c_{j,t+1} + P_{j,t+1}])$ imbedded in date t-1 price.
- Now consider $\text{cov}_{t-1}(MRS_{i,t-1,t}, P_{j,t} + c_{j,t})$. Note time subscripts and no expectation E_t inside here.
- Different $MRS_{j,t-1,t}$ for different future states at future date t.
 - In some states at date t, marginal investor has high marginal valuation $V'_{i,t}$ for additional wealth (i.e., $MRS_{j,t-1,t}$ is high at t-1).
 - In others states at date t, $V'_{i,t}$ is smaller (but still positive) (i.e., $MDF_{j,t-1,t}$ is low).
- Consider 3 different securities with **same future payoff expectations** $E_{t-1}(c_{j,t} + P_{j,t})$ but **different covariances** of their future payoffs with the marginal investor's $MRS_{j,t-1,t}$
- Which will have a higher value today?



Why are some factors priced? (2)

$$P_{j,t-1} = D_{t-1} E_{t-1}[P_{j,t} + c_{j,t}] + \text{cov}_{t-1}(\text{MRS}_{i,t-1,t}, P_{j,t} + c_{j,t}).$$

- **Question:** What types of securities j is Investor i willing to pay a high price for?
 - **Expected next period's payoff** $E_{t-1}[P_{j,t} + c_{j,t}]$ is **high**.
 - Randomness in next period payoff $P_{j,t} + c_{j,t}$ **covaries positively** with i 's future $\text{MRS}_{i,t-1,t}$.
 - Such securities give more wealth in “states” in which i values wealth more highly.
- **Question:** Why is investor i 's $\text{MRS}_{i,t-1,t}$ a “random variable”? Because it depends on ...
 - Random **investment wealth** $\sum_{j=1}^N q_{i,j,t-1} * (P_{j,t} + c_{j,t})$ at time t given random prices & dividends
 - Random **non-investment income** $L_{i,t}$ for i at time t .
 - Random **other things** $x_{i,t}$ investor cares about at time t :
 - **Investment opportunities** at time t : $E_t[\text{ret}]$, variances, factor loadings of traded assets, etc.
 - Anything else that affects investor's preferences at time t – e.g., investor sentiment.

Why are some factors priced? (3)

Examples

- Tricky part: $MRS_{i,t-1,t}$ may be **decreasing** in some factors but increasing in other factors.
- If investor i is **risk averse**, then, marginal value $V'_{i,t}$ of additional \$1 (and, thus, $MRS_{i,t-1,t}$) is low (high) when her total wealth $W_{i,t}$ at time t is high (low)
 - Risk-averse investors pay more at $t-1$ for securities that are “**wealth hedges**” in that a security’s date t payoff $P_{j,t} + c_{j,t}$ covaries **negatively** with investor’s future wealth and, thus, **positively** with their $MRS_{i,t-1,t}$.
- Impact of better **investment opportunities** in $x_{i,t}$ is complex. Might make $MRS_{i,t-1,t}$ larger (since investing at t is more productive) or smaller (since given better investment conditions, don’t need as much wealth to fund future consumption).
 - Impact of market conditions $x_{i,t}$ on $V'_{i,t}$ and, thus, on $MRS_{i,t-1,t}$ is ambiguous

Punch line

- Variables that induce randomness in marginal investor $MRS_{i,t-1,t}$ that covaries with randomness in asset payoffs $P_{j,t} + c_{j,t}$ at time t will be priced at $t-1$.
 - If marginal investor holds well-diversified portfolio, then $MRS_{i,t-1,t}$ driven by systematic factor randomness. Thus, systemic factors should be priced, but not idiosyncratic randomness!

Why are some factors priced? (4)

Markets

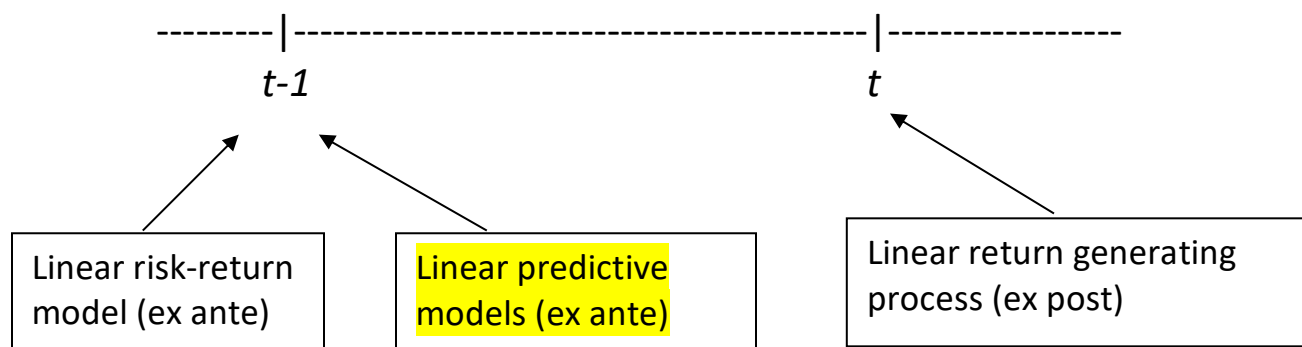
- ❑ Market consists of a bunch of people solving investment-optimization problems
- ❑ Common factors in prices P , cash distributions c & other income L and investor preferences are systematic risks that affect large numbers of investors
- ❑ Investors must collectively hold enough of each stock so that markets clear
- ❑ Thus, systemic exposures can be priced.

Equilibrium

- ❑ Pricing depends on MRS for **marginal investor** in aggregate market.
- ❑ In classic asset pricing models (e.g., Lucas, Merton), **every** investor is marginal. Thus, can use any investor i 's MRS to price securities.
- ❑ More complicated in the real world. In reality, only **some** investors are marginal. **Who?**

Linear models yet again

Timing:



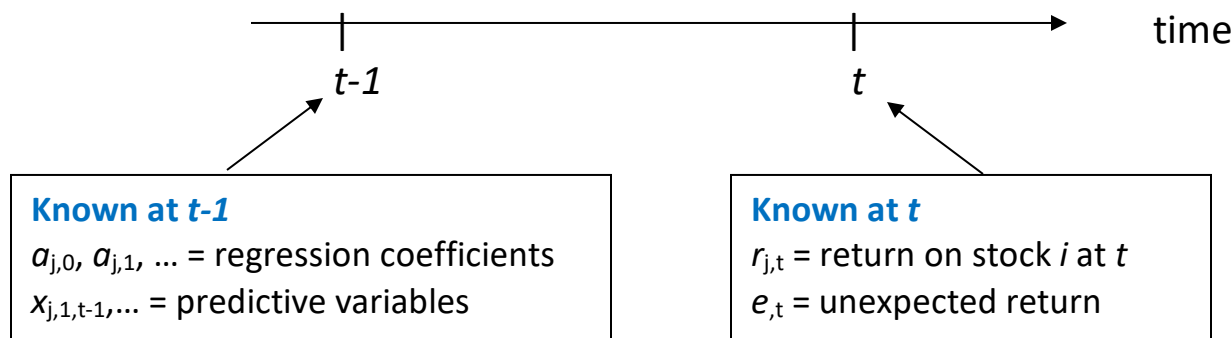
Linear predictive models

- ❑ Predicts returns r_{jt} at t using prior explanatory variables known at $t-1$.
- ❑ Used for security selection for individual securities
- ❑ Also used for market timing across asset classes

Linear models of risk/reward

- ❑ Relates expected returns $E_t[r_{jt}]$ given date $t-1$ prices to risk measures known at $t-1$.

Predictive linear models



Returns and prediction

$$r_{j,t} = E_{t-1}[r_{j,t} \mid \text{everything known at } t-1] + e_{j,t}$$

$$= a_0 + a_1 x_{j,1,t-1} + \dots + a_K x_{j,K,t-1} + e_{j,t}$$

- **Ex ante predictive** model. Uses date $t-1$ variables to predict date t returns.
- Does not assume expected returns are just compensation for risk.
- Expected returns may also include **non-risk asset characteristics** and possible **market mispricing**.
- Explanatory variables $x_{j,1,t-1}, x_{j,2,t-1}, \dots$ are **asset characteristics**.
 - Can be risk sensitivities (e.g., factor loadings, idiosyncratic SE). But not limited to risk metrics.
 - Can be other characteristics (firm size, firm B/M, etc) known about firms at time $t-1$
 - Can be market conditions that predict returns (market sentiment, market liquidity, errors)
- Predictive coefficients may change over time ($a_{k,t}$) rather than being constants (a_k).

Fama-MacBeth estimation

Goal: To use panel (firm-time) data to estimate expected-return premia $\{a_{0,t}, \dots, a_{K,t}\}$ or $\{a_0, \dots, a_K\}$

Estimation strategy

- **Panel data:** Returns and firm characteristics for $J = 1, \dots, N$ stocks and $t = 1, \dots, T$ months
- For each month t , run a **cross-sectional regression** of end-of-month returns on beginning-of-month asset characteristics across stocks $j = 1, \dots, N$.

$$r_{j,t} = a_{0,t} + a_{1,t} x_{1,j,t-1} + \dots + a_{K,t} x_{K,j,t-1} + e_{j,t}$$

→ Time series of estimated coefficients $\{\hat{\alpha}_{0,1}, \dots, \hat{\alpha}_{0,T}\}, \dots, \{\hat{\alpha}_{K,1}, \dots, \hat{\alpha}_{K,T}\}$ from regressions for months $t = 1, \dots, T$

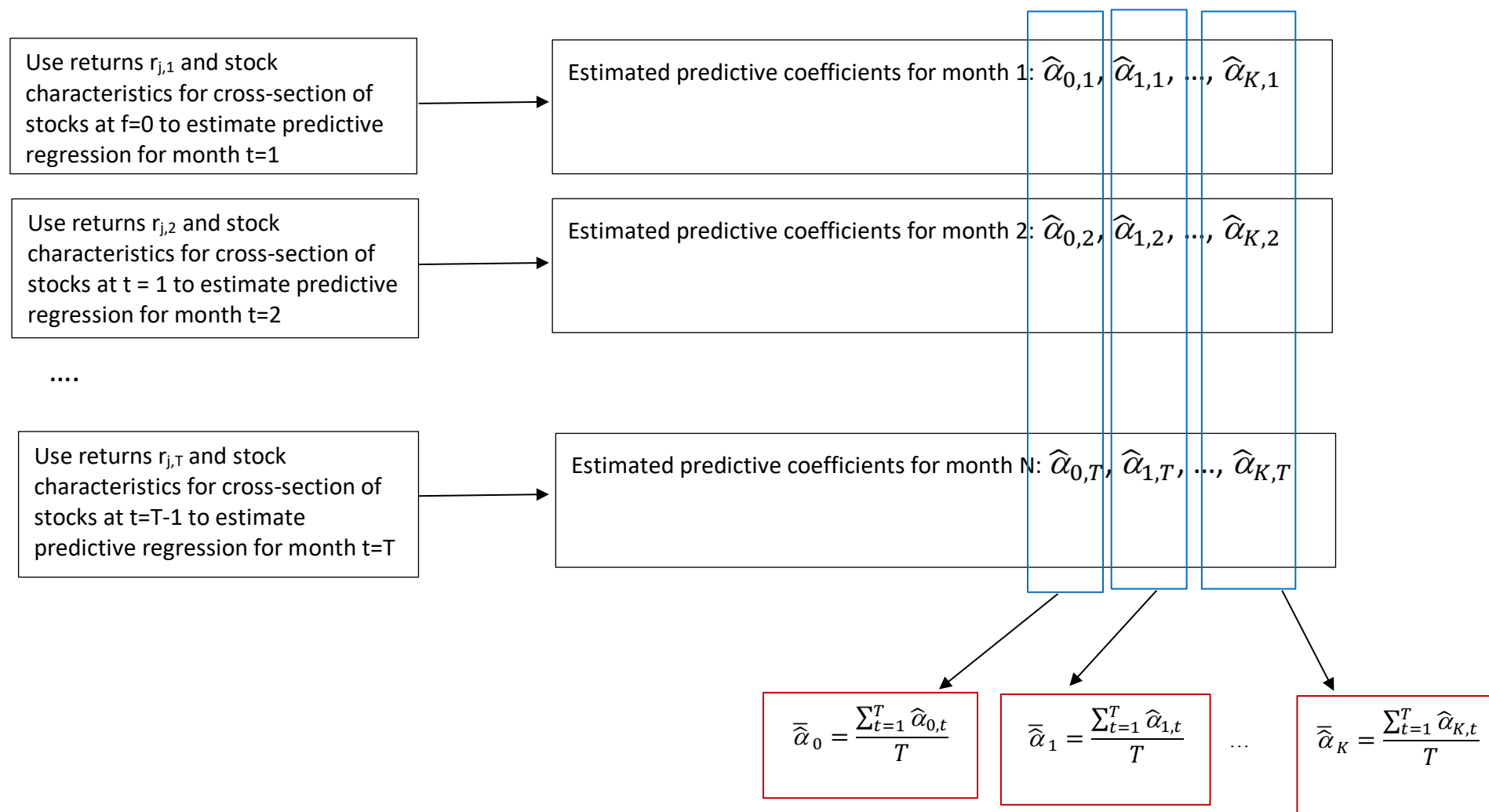
- **Averages:** One possibility is to average estimated predictive coefficients over all N periods to estimate the **unconditional** predictive coefficients:

$$\bar{\alpha}_0 = \frac{\sum_{t=1}^T \hat{\alpha}_{0,t}}{T}, \dots, \bar{\alpha}_K = \frac{\sum_{t=1}^T \hat{\alpha}_{K,t}}{T}$$

- **Rolling averages:** Another possibility is to average estimated coefficients over prior rolling windows (e.g., 12 months) to estimate changing **conditional** predictive coefficients at time t :

$$\bar{\alpha}_{0,t} = \frac{\sum_{s=t-12}^t \hat{\alpha}_{0,s}}{12}, \dots, \bar{\alpha}_{K,t} = \frac{\sum_{s=t-12}^t \hat{\alpha}_{K,s}}{12}$$

Schematic for FM estimates of unconditional predictive coefficients



Example: Is liquidity priced?

- **Paper:** Illiquidity and Stock Returns: Cross-Section and Time-Series Effects, Y. Amihud, *Journal of Financial Markets*, 5 (2002) 31-56

- Measure of illiquidity: $ILLIQ_{j,t} = \frac{\sum_{d=1}^{D_t} |r_{j,t,d}| / vol_{j,t,d}}{D_t}$ = average across D_t days in year t

- $ILLIQMA_{j,t} = ILLIQ_{j,t} / [\text{cross-sectional average stock ILLIQ in year } t]$

Table 2

Cross-section regressions of stock return on illiquidity and other stock characteristics^a

The table presents the means of the coefficients from the monthly cross-sectional regression of stock return on the respective variables. In each month of year y , $y = 1964, 1965, \dots, 1997$, stock returns are regressed cross-sectionally on stock characteristics that are calculated from data in year $y - 1$. *BETA* is the slope coefficient from an annual time-series regression of daily return on one of 10 size portfolios on the market return (equally weighted), using the Scholes and Williams (1977) method. The stock's *BETA* is the beta of the size portfolio to which it belongs. The illiquidity measure *ILLIQ* is the average over the year of the daily ratio of the stock's absolute return to its dollar trading volume. *ILLIQ* is averaged every year across stocks, and *ILLIQMA* is the respective mean-adjusted variables, calculated as the ratio of the variable to its annual mean across stocks (thus the means of all years are 1). In *SIZE* is the logarithm of the market capitalization of the stock at the end of the year, *SDRET* is the standard deviation of the stock daily return during the year, and *DIVYLD* is the dividend yield, the sum of the annual cash dividend divided by the end-of-year price. *R100* is the stock return over the last 100 days and *R100YR* is the return during the period between the beginning of the year and 100 days before its end.

The data include 408 months over 34 years, 1964–1997, (the stock characteristics are calculated for the years 1963–1996). Stocks admitted have more than 200 days of data for the calculation of the characteristics in year $y - 1$ and their end-of-year price exceeds \$5. Excluded are stocks whose *ILLIQ* is at the extreme 1% upper and lower tails of the respective distribution for the year.

Variable	All months	Excl. January	1964–1980	1981–1997	All months	Excl. January	1964–1980	1981–1997
<i>Constant</i>	−0.364 (0.76)	−0.235 (0.50)	−0.904 (1.39)	0.177 (0.25)	1.922 (4.06)	1.568 (3.32)	2.074 (2.63)	1.770 (3.35)
<i>BETA</i>	1.183 (2.45)	0.816 (1.75)	1.450 (1.83)	0.917 (1.66)	0.217 (0.64)	0.260 (0.79)	0.297 (0.59)	0.137 (0.30)
<i>ILLIQMA</i>	0.162 (6.55)	0.126 (5.30)	0.216 (4.87)	0.108 (5.05)	0.112 (5.39)	0.103 (4.91)	0.135 (3.69)	0.088 (4.56)
<i>R100</i>	1.023 (3.83)	1.514 (6.17)	0.974 (2.47)	1.082 (2.96)	0.888 (3.70)	1.335 (6.19)	0.813 (2.33)	0.962 (2.92)
<i>R100YR</i>	0.382 (2.98)	0.475 (3.70)	0.485 (2.55)	0.279 (1.59)	0.359 (3.40)	0.439 (4.27)	0.324 (2.04)	0.395 (2.82)
<i>Ln SIZE</i>					−0.134 (3.50)	−0.073 (2.00)	−0.217 (3.51)	−0.051 (1.14)
<i>SDRET</i>					−0.179 (1.90)	−0.274 (2.89)	−0.136 (0.96)	−0.223 (1.77)
<i>DIVYLD</i>					−0.048 (3.36)	−0.063 (4.28)	−0.075 (2.81)	−0.021 (2.11)

^a *t*-statistics in parentheses.

Haugen and Baker (1996)

Paper: “Commonality in the determinants of expected stock returns” by R. Haugen and N. Baker, *Journal of Financial Economics* 41 (1996) 401-439

- Example of trading strategy using Fama-MacBeth predictive coefficient estimates
- Given current rolling average estimated predictive coefficients in month t , plug in the current predictive variables for each stock $j = 1, \dots, N$ at time t to **get predicted returns** for month $t+1$

$$\hat{E}_t[r_{j,t+1}] = \bar{\alpha}_{0,t} + \bar{\alpha}_{1,t}x_{1,j,t} + \dots + \bar{\alpha}_{K,t}x_{K,j,t}$$

- **Rank** stocks from highest to lowest predicted returns.
- **Long-short** trading strategy
 - Go long \$1 of a portfolio of top 10% of predicted “winners.”
 - Short \$1 of a portfolio of bottom 10% predicted “losers.”
 - Hold one month. Update rolling estimated coefficients. Repeat for next month.
 - Trying to predict which stocks will outperform others each month

Adaptive strategy:

- Intended to capture **systematic mispricing** of **firm characteristics across stocks**.
- Assumes **consistent market mispricing** of characteristics across different stocks.
- Rolling coefficients allow predictive relationships to **evolve** over time.
 - Important since market has a strong incentive to correct perceptual biases over time.
 - Mispricing is **endogenous, not exogenous**. Psychology, not physics.

H&B Results

- R^2 from monthly cross-sectional regressions

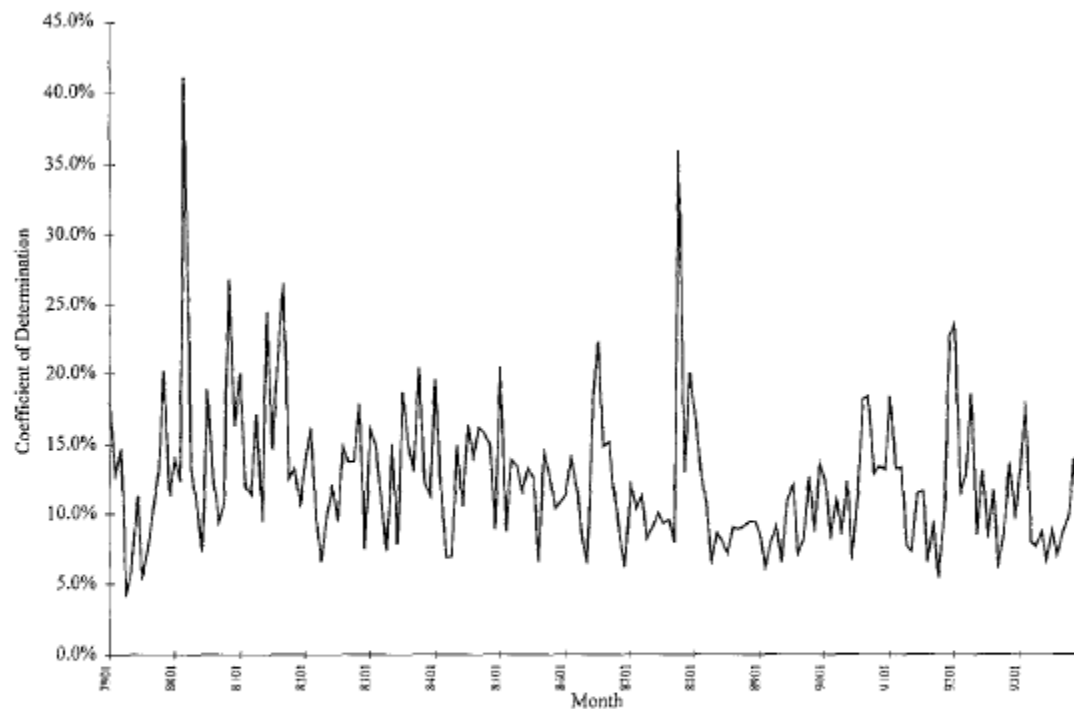


Fig. 1. Time series of adjusted R -squareds for the expected return factor model.

In each month, over the period 1979 through 1993, the cross-section of returns to the stocks in the Russell 3000 stock index are regressed (using OLS) on the cross-section of various firm characteristics related to risk, liquidity, price level, growth potential, and price history. The time series of the adjusted coefficients of determination for the individual monthly regressions are shown in the figure.

H&B Results (2)

- Most important monthly explanatory variables based on t-statistics

Table 1

The mean and *t*-statistics for the most important factors of 1979/01 through 1986/06

Factor	1979/01 through 1986/06		1986/07 through 1993/12	
	Mean	<i>t</i> -stat.	Mean	<i>t</i> -stat.
One-month excess return	−0.97%	−17.04	−0.72%	−11.04
Twelve-month excess return	0.52%	7.09	0.52%	7.09
Trading volume/market cap	−0.35%	−5.28	−0.20%	−2.33
Two-month excess return	−0.20%	−4.97	−0.11%	−2.37
Earnings to price	0.27%	4.56	0.26%	4.42
Return on equity	0.24%	4.34	0.13%	2.06
Book to price	0.35%	3.90	0.39%	6.72
Trading volume trend	−0.10%	−3.17	−0.09%	−2.58
Six-month excess return	0.24%	3.01	0.19%	2.55
Cash flow to price	0.13%	2.64	0.26%	4.42
Variability in cash flow to price	−0.11%	−2.55	−0.15%	−3.38

- Lagged returns
- Accounting variables: Book-to-Market, EPS
- Market volume either as a measure of liquidity or as a metric for fads
- Not many explicit “risk” measures

H&B Results (3)

- Statistics for rebalanced “decile” portfolios.
- Seems to be a large spread in average returns

Table 2

U.S. realized annualized returns across deciles 1 to 10 formed by ranking on expected return

	Dec. 1	Dec. 2	Dec. 3	Dec. 4	Dec. 5	Dec. 6	Dec. 7	Dec. 8	Dec. 9	Dec. 10	Slope	R ²
Annual return												
1979	33.5%	32.6%	33.9%	43.1%	35.2%	36.3%	47.3%	40.1%	39.3%	43.4%	1.1%	0.446
1980	17.4%	26.2%	25.4%	27.2%	25.8%	41.3%	42.6%	45.3%	55.6%	68.4%	5.0%	0.897
1981	−15.6%	−14.2%	−7.9%	−4.6%	2.1%	5.6%	0.4%	6.3%	9.7%	16.2%	3.3%	0.931
1982	3.2%	15.5%	21.8%	24.6%	24.0%	25.9%	32.1%	34.6%	39.5%	49.7%	4.1%	0.929
1983	11.8%	18.0%	23.4%	29.5%	28.8%	39.3%	37.8%	46.1%	45.1%	54.5%	4.4%	0.962
1984	−30.9%	−20.7%	−13.4%	−9.1%	−6.5%	1.0%	2.8%	12.8%	15.4%	22.4%	5.5%	0.986
1985	4.3%	18.4%	26.6%	37.8%	34.9%	37.8%	34.9%	41.2%	43.4%	45.7%	3.7%	0.776
1986	−15.2%	−7.1%	1.9%	9.2%	12.1%	15.1%	19.9%	23.2%	23.0%	30.9%	4.7%	0.925
1987	−23.8%	−12.3%	−5.0%	−6.8%	0.0%	1.6%	−3.4%	−2.0%	1.2%	−5.1%	1.8%	0.486
1988	1.5%	10.4%	18.5%	24.0%	22.2%	28.8%	26.9%	25.9%	29.7%	27.0%	2.5%	0.714
1989	−3.0%	8.2%	9.7%	16.8%	18.7%	21.5%	28.5%	29.8%	32.4%	28.7%	3.6%	0.893
1990	−46.9%	−36.2%	−27.5%	−21.7%	−15.5%	−12.7%	−10.2%	−9.9%	−2.9%	1.3%	4.8%	0.937
1991	23.9%	29.3%	36.5%	42.0%	45.2%	45.7%	51.1%	46.6%	46.9%	57.4%	3.1%	0.817
1992	2.5%	7.5%	16.3%	20.3%	17.8%	15.7%	17.1%	18.9%	21.1%	24.5%	1.8%	0.619
1993	6.4%	9.2%	18.2%	18.5%	19.9%	20.1%	20.0%	20.7%	24.2%	22.2%	1.6%	0.738
Average returns												
1979–93	−4.5%	3.7%	10.3%	15.1%	16.5%	22.3%	21.6%	24.0%	27.1%	30.9%	3.5%	0.932
Annualized risk												
1979–93	22.62%	20.59%	19.28%	19.21%	18.19%	18.10%	17.83%	17.95%	17.45%	18.50%		

The 3000 stocks in the Russell 3000 stock index are ranked and formed into equally weighted deciles on the basis of their relative expected monthly returns. At the beginning of each month, the expected returns are computed by summing the products of (a) the individual factor exposures for each stock and (b) the average of the individual factor payoffs over the previous 12 months. The factor exposures are based on firm characteristics related to risk, liquidity, price level, growth potential, and price history. The deciles are re-formed monthly. The realized monthly returns to the deciles are then linked, and the annual returns and the averages of annual returns are reported in the table. The slopes and coefficients of determinations are obtained by regressing the yearly and average realized returns on the decile rankings. The volatilities are the annualized standard deviations of the monthly rates of return for each decile.

H&B Results (4)

- Predicted winners don't look riskier than predicted losers

Table 3

The values for selected average firm characteristics by decile (1 = low expected return, 10 = high expected return)

	Dec. 1	Dec. 2	Dec. 3	Dec. 4	Dec. 5	Dec. 6	Dec. 7	Dec. 8	Dec. 9	Dec. 10
Risk										
Market beta	1.21	1.16	1.13	1.11	1.09	1.07	1.05	1.03	1.02	1.00
Volatility (total return)	41.42%	38.42%	36.99%	36.00%	35.16%	34.29%	33.59%	32.86%	32.50%	33.22%
Debt to equity	1.03	0.95	0.89	0.87	0.85	0.83	0.83	0.82	0.85	0.85
Debt to equity growth	0.27%	0.12%	0.08%	0.05%	0.05%	0.06%	0.05%	0.02%	0.04%	-0.03%
Interest coverage	1.76	4.63	5.74	6.36	6.48	6.66	6.98	6.98	6.98	6.63
Interest coverage growth	-0.64%	-0.31%	-0.19%	-0.10%	-0.12%	-0.11%	-0.09%	-0.05%	-0.05%	-0.02%
Liquidity										
Trading volume (millions/month)	\$42.42	\$42.42	\$47.19	\$42.74	\$65.23	\$51.79	\$56.02	\$51.73	\$60.94	\$60.89
Market capitalization (millions)	\$470	\$513	\$564	\$593	\$635	\$680	\$755	\$843	\$931	\$1011
Price per share	\$14.93	\$18.03	19.91	\$21.21	\$22.58	\$24.01	\$25/58	\$27.62	\$29.31	\$30.21

Haugen and Baker Summary

Punchline: Back-testing results show

- ❑ Strong predictability in relative returns across stocks.
 - ❑ High returns on apparently low risk portfolios.
- Looks like a portfolio manager's dream.

Questions

- ❑ Why is apparent performance of long/short portfolios so spectacular given mediocre R^2 s?
- ❑ Other questions?

Other comments

- ❑ H&B illustrates type of quantitative security-selection strategy used by long-short hedge funds.
- ❑ H&B say their results “disprove” Efficient Markets. But not everyone is convinced.
- ❑ What is your initial reaction?

Learning objectives

You should be able to ...

- ❑ Compute linear relation between risk sensitivities and expected returns in CAPM and APT
- ❑ Explain conceptually what makes something priced?
- ❑ Do Fama-MacBeth estimation and able to interpret results
- ❑ Test if a variable is priced using Fama-MacBeth as in priced liquidity example
- ❑ Set up a long-short trading strategy
- ❑ Set up a back-testing analysis